

Yen-Hsiang Chang

Graduate Student Researcher, UC Berkeley

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Research Interests	High-Performance Computing, Parallel Graph Algorithms, Scalable Sparse Solvers, Applied Numerical Linear Algebra	
Education	University of California, Berkeley	Berkeley, CA
	Ph.D. Candidate in EECS Advisor: Prof. James Demmel	2023 – Present
	University of Illinois Urbana-Champaign	Champaign, IL
	B.S. in Grainger Engineering Major: Computer Engineering, Minor: Mathematics Cumulative GPA: 3.99/4.00, Major GPA: 4.00/4.00, Minor GPA: 4.00/4.00 Graduated with Highest Honors, on completion of a thesis of superior quality.	2018 – 2022
Publications	Parallelizing the Approximate Minimum Degree Ordering Algorithm: Strategies and Evaluation	
	Yen-Hsiang Chang, Aydın Buluç, James Demmel <i>Proceedings of the SIAM Conference on Parallel Processing for Scientific Computing (PP)</i> , 2026. Forthcoming.	
	Parallelizing Maximal Clique Enumeration on GPUs	
	Mohammad Almasri*, <u>Yen-Hsiang Chang*</u> , Izzat El Hajj, Rakesh Nagi, Jinjun Xiong, Wen-mei Hwu <i>International Conference on Parallel Architectures and Compilation Techniques (PACT)</i> , 2023.	
	MLHarness: A Scalable Benchmarking System for MLCommons	
	Yen-Hsiang Chang, Jianhao Pu, Wen-mei Hwu, Jinjun Xiong <i>BenchCouncil Transactions on Benchmarks, Standards and Evaluations, (TBench)</i> , 2021.	
Research Experiences	Sparsitute, PASSION Lab and BeBOP	Berkeley, CA
	Graduate Researcher, UC Berkeley and LBNL Mentors: Prof. James Demmel, Dr. Aydın Buluç Focus: Scalable direct sparse solvers and fill-reduced ordering algorithms	Aug 2023 – Present
	Coordinated Science Laboratory	Champaign, IL
	Undergraduate Researcher, UIUC Mentors: Prof. Wen-mei Hwu, Prof. Rakesh Nagi, Prof. Jinjun Xiong Focus: Parallel graph mining on GPUs	May 2021 – May 2022
	Center for Cognitive Computing Systems Research	
	Undergraduate Researcher, UIUC Mentors: Prof. Wen-mei Hwu, Prof. Jinjun Xiong Focus: Benchmarking methods for evaluating ML inference systems	
Work Experiences	NVIDIA Corporation	Santa Clara, CA
	Linear Algebra Libraries Intern	May 2026 – Aug 2026

Teaching Experiences	Teaching assistant, University of California, Berkeley	
	CS 170: Efficient Algorithms and Intractable Problems CS 267: Applications of Parallel Computers	Fall 2026 Spring 2025
Selected Talks	SIAM Conference on Parallel Processing for Scientific Computing (PP26)	Berlin, Germany Mar 3, 2026
	Title: Parallelizing the Approximate Minimum Degree Ordering Algorithm: Strategies and Evaluation	
	SIAM Conference on Applied and Computational Discrete Algorithms (ACDA25)	Montréal, Canada Jul 31, 2025
	Title: Parallelizing the Approximate Minimum Degree Ordering Algorithm: Strategies and Evaluation (Extended Abstract) Best Presentation Award	
	The 32nd International Conference on Parallel Architectures and Compilation Techniques (PACT23)	Vienna, Austria Oct 24, 2023
	Title: Parallelizing Maximal Clique Enumeration on GPUs	
Skills	C/C++, Python, CUDA, OpenMP, MPI	
Awards	Best Presentation Award in SIAM ACDA25	2025
	Tong Leong Lim Pre-Doctoral Prize Highest distinction in the pre-doctoral examination in EECS at UC Berkeley	2025
	17th Place in 2022 Google Hash Code World Finals	2022
	Bronze Medal in 2020 ICPC World Finals 11th place worldwide	2021
	ECE Alumni Association Scholarship Outstanding scholastic record in ECE at UIUC	2021
	163rd Place in 2021 Google Code Jam Round 3	2021
	6th Place in Microsoft Q# Summer Coding Contest	2020
	Round 4 Qualifier in 2020 Topcoder Open Algorithm Competition Top 110 contestants worldwide	2020
	Midwest Champion in 2020 ICPC North America Championship 10th place nationwide	2020
	2nd Place in 2020 ICPC North America Championship Cyber Challenge	2020
	132nd Place in 2020 Google Code Jam Round 3	2020
	1st Place in 2019 ICPC North America Mid-Central Regional Contest	2019
	112th Place in 2019 Google Code Jam Round 3	2019
	Dean's List in Grainger College of Engineering at UIUC	2018 – 2022